



Botany, traditional use, phytochemistry, pharmacology and toxicology of *Sigesbeckiae Herba* (*Xixiancao*): a review

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Received: 28 February 2019 / Accepted: 6 January 2020 / Published online: 11 August 2020
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Abstract *Sigesbeckiae Herba* is a traditional Chinese herbal medicine and has been used for inflammatory diseases with a long history. The current official plant origins of *Sigesbeckiae Herba* include *Sigesbeckia orientalis* L. (SO), *S. pubescens* Markino (SP) and *S. glabrescens* Markino (SG). In this work, a systemic review about the scientific work for medicinal *Sigesbeckiae Herba* regarding their botanical descriptions, reported bioactive components, pharmacological activities and toxicities was performed. The results indicate that a total of 251 compounds including flavonoids, diterpenoids, sesquiterpenoids, triterpenoids, and sterols have been isolated from *Sigesbeckia* plants and reported to be

characteristically distributed in SO, SP and SG. The extracts and pure compounds derived from *Sigesbeckiae Herba* are reported to present various pharmacological activities including anti-inflammation, immunomodulation, anti-cancer, and others. Although no significant side-effects and toxicities for *Sigesbeckiae Herba* have been reported in clinics, the potential pulmonary toxicity for the water extracts of SH are reported in experimental animals. In conclusion, *Sigesbeckiae Herba* is a valuable Chinese herbal medicine and has been investigated to present multiple pharmacological activities. The main pharmacological investigations support its traditional use for treating chronic inflammatory diseases. The available literature shows that much of the activity of *Sigesbeckiae Herba* can be attributed to diterpenoids, sesquiterpenoids and flavonoids. However, further investigations on the molecular mechanisms to pharmacological activities and the potential toxicities is recommended. Moreover, further investigations on the comparative comprehension among the three medicinal *Sigesbeckia* plants (SO, SP and SG) both in chemistry and pharmacology are also recommended for their precise applications in clinics.

Electronic supplementary material The online version of this article (<https://doi.org/10.1007/s11101-020-09714-4>) contains supplementary material, which is available to authorized users.

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Keywords *Sigesbeckiae Herba* · Traditional use ·
Phytochemistry · Pharmacology · Toxicology

Introduction

Sigesbeckiae Herba (SH, also called *Xixiancao* in Chinese) is a traditional anti-inflammatory Chinese herbal medicine and has been widely used for treatment of various inflammatory diseases. Historically, SH was firstly recorded in the ancient Chinese medicinal book of “*Newly revised materia medica*” (the Tang Dynasty), and further developed and evolved as recorded in subsequent ancient Chinese medicinal books including *Classified Materia Medica from Historical Classics for Emergency* (the Northern Song Dynasty), *Herbal for Relief of Famines* (the Ming Dynasty) and *Compendium of Materia Medica* (the Ming Dynasty). The plants ever used for SH included *Sigesbeckia pubescens* Markino (family of *Compositae*) (Fig. 1b), *Solanum nigrum* L. (family of *Solanaceae*) and *Carpesium abrotanoides* L. (family of *Compositae*). However, based on a textual research of herbal materia medica, only *S. pubescens* Markino (SP) was considered as the correct and historical used plant for SH. In 1963, SP was firstly recorded in the Chinese Pharmacopeia and recommended to be used as SH; while after 1977, other two *Sigesbeckia* species, *S. orientalis* L. (SO) (Fig. 1a), and *S. glabrescens* Markino (SG) (Fig. 1c) were also recorded in the Chinese Pharmacopeia and recommended to be used as SH. Currently, the plant origins of commercial medicinal material of SH include SO, SP and SG.

In the past decades, a numerus of studies has been performed on the chemical, pharmacological and

toxicological properties of SH. Among the three species, SP is the mostly studied one, thereafter the SO and SG. Chemically, the reported components in SH include terpenoids (sesquiterpenoids, diterpenoids and triterpenoids), flavonoids, sterols and some others. Terpenoids are considered as the most important ingredients for SH which related to their clinical therapeutic effects. Pharmacologically, SH has been reported to present various pharmacological activities including anti-inflammatory (Hong et al. 2014; Jeon et al. 2014; Wang et al. 2008), anti-allergy (Hwang et al. 2001; Kim et al. 2001), anti-thrombotic, anti-atherosclerosis, improving microcirculation (Lee et al. 2013; Wang et al. 2011a, b, c, d), promoting skin wound healing (Sung et al. 2011; Wang et al. 2011a), and anti-cancer (Chang et al. 2014; Lee et al. 2014a, b; Cho et al. 2013; Jun et al. 2006) activities. Moreover, the water extract of SH was reported to induce pulmonary toxicity in experimental animal (Guan et al. 2008). However, there is still no systematic review summarizes and prospects the progress of SH-related researches yet. The aim of the present review is to compile an up-to-2018 and comprehensive knowledge of three mainstream SH species by comparing their botanic, chemistry, pharmacology and toxicology, and to provide the potential tendency and perspectives for future research of the multi-origin and traditional Chinese herbal medicine of SH.

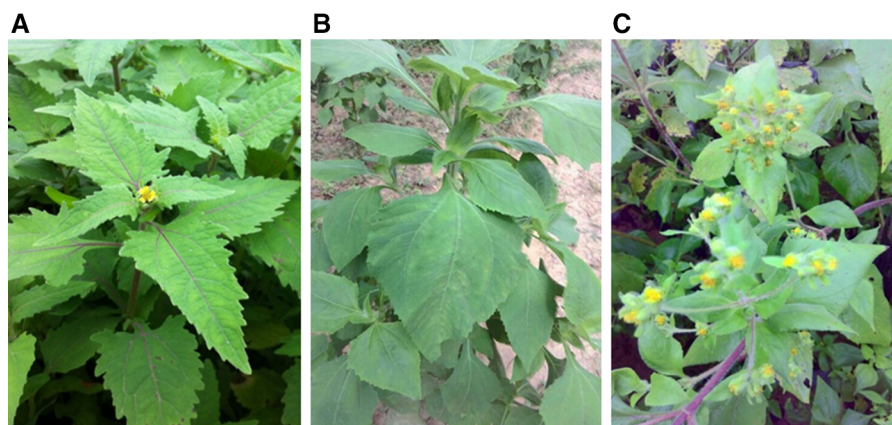


Fig. 1 The plants of *Sigesbeckia orientalis* L. (SO) (a), *S. pubescens* Markino (b) and *S. glabrescens* Markino (SG) (c)

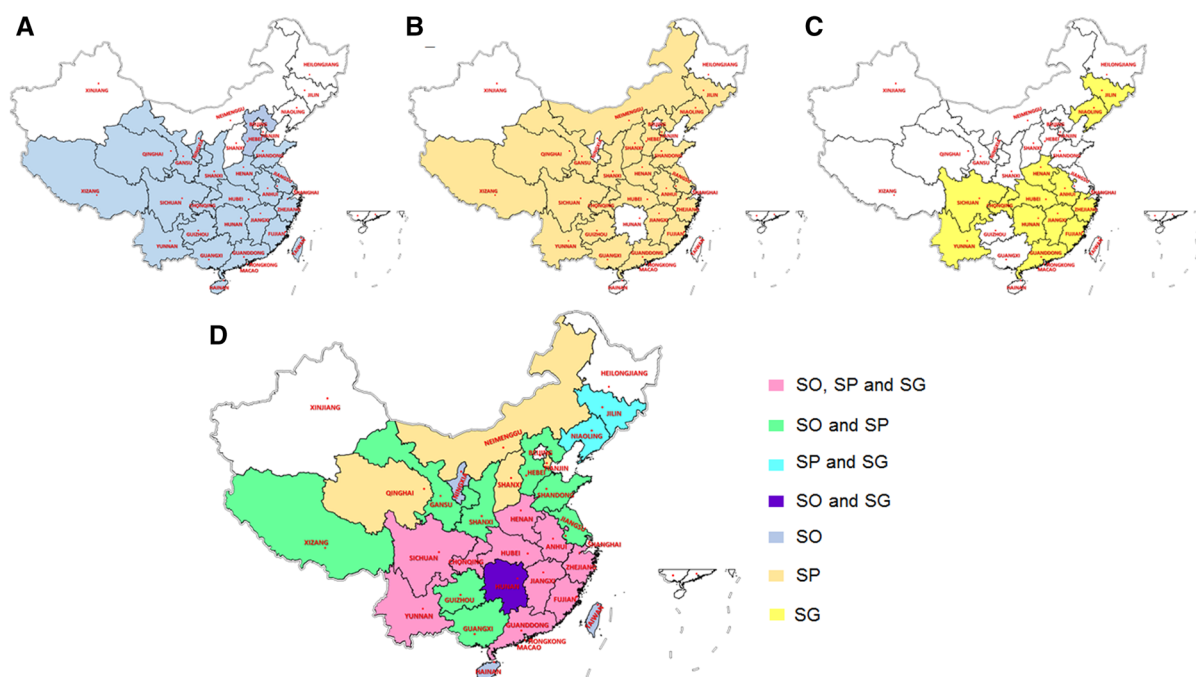


Fig. 2 The geographical distribution characteristics of *Sigesbeckia orientalis* L. (SO) (a), *S. pubescens* Markino (b), *S. glabrescens* Markino (SG) (c) and their overlapped distribution (d)

Botanical description and distribution

Sigesbeckia plants are found to be extensively distributed in China, Japan, Korea and Southeast Asia. The individual and overlapped (Fig. 2d) distributions (Wu and Raven 2011) of SO (Fig. 2a), SP (Fig. 2b) and SG (Fig. 2c) geographically in China were illustrated in Fig. 2. As shown in the Figures, all three *Sigesbeckia* species could be found in East China, Central China, Southwest China and South China. Northwest China might be specific to SO and SP, while Northeast China might be specific to SP and SG. Moreover, the main species distributed in North China is SP. The main botanical descriptions of SO, SP and SG are summarized in Table 1.

Local and traditional uses

SH has been widely used in China and some other Asian countries. Traditionally in China, the aerial part of the plant is used for medicinal purposes, and the best harvest time for SH is from April to October (i.e. the lush period of pre-florescence or florescence)

depending on the individual species and its growth regions (Chou et al. 2006). SH is used to improve the clinical symptoms of wind-dampness arthralgia and pain, hypodynamia of sinew and bone, soreness and weakness of waist and knees, numbness of the limbs, hemiplegia, urticaria and eczema sore by dispelling wind-dampness, soothing painful joints and detoxifying. Due to the characteristics of SH described by Chinese medical theories (bitter in taste, cold in property, and with small toxicity), processing with rice wine (nine-steam-nine-bask) is considered to be an effective approach to decrease the side-effects of SH for its long-term applications (Gao et al. 2008; Guan et al. 2007, 2005). However, the changes in chemical profile and pharmacological activities of SH before and after wine-steam was only limited investigated and reported until now (Su et al. 2014; Kong et al. 2014).

In China, the pharmaceutical preparations derived from SH alone (Xixian pills/capsule) or SH-contained herbal formulas (Xitong pills/capsule, Xihong Tongluo oral solution, Xixian Tongshuan capsule, etc.) has been widely used in clinic for the management of

Table 1 The specific botanical and morphological characteristics of *Sigesbeckia orientalis* L (SO), *S. pubescens* Makino (SP), and *S. glabrescens* Makino (SG)

Plant species	Environment	Morphological specificity		
		Stem	Leaves	Flowers
<i>S. orientalis</i> L	Anhui, Fujian, Gansu, Guangdong, Guangxi, Guizhou, Hainan, Henan, Hubei, Hunan, Jiangsu, Jiangxi, Shaanxi, Sichuan, Taiwan, Xizang, Yunnan, Zhejiang	30–100 cm tall; erect, simple or dichotomously branched, branches opposite; covered with densely pubescent, especially in upper part	Ovate-triangular, ovate, or oblong-ovate; with finely and appressed pilose, especially abaxially; margin largely and usually unequally toothed to sinuate-dentate	Capitula; densely pubescent on the peduncle; without or with few stipitate-glandular hairs Florescence: April to September
<i>S. pubescens</i> Markino	Anhui, Fujian, Gansu, Guangdong, Guangxi, Guizhou, Hainan, Hebei, Henan, Hubei, Hunan, Jilin, Liaoning, Jiangsu, Jiangxi, Nei Mongol, Shaanxi, Sichuan, Taiwan, Xizang, Yunnan, Zhejiang	60–120 cm tall; covered with densely white pubescent, especially on upper part	Median cauline leaves with winged petiole; blade ovate-deltate, regularly toothed; short appressed pubescence on both surfaces; veins on lower surface with densely white pubescent	Capitula radiate; peduncle with densely glandular pilose or long pilose Florescence: May to August
<i>S. glabrescens</i> Markino	Anhui, Fujian, Guangdong, Guangxi, Guizhou, Hainan, Henan, Hubei, Hunan, Jiangsu, Jiangxi, Liaoning, Sichuan, Taiwan, Yunnan, Zhejiang	35–100 cm tall; with shortly appressed pilose	Median cauline leaves with winged petiole; blade ovate-deltate, regularly toothed; shortly appressed pubescence on upper surface; upper leaves sessile, oblong; uppermost leaves linear	Capitula radiate; peduncle without glandular pilose Florescence: April to September

various chronic inflammatory diseases such as rheumatoid arthritis.

Based on the traditional use, in 2015, Phynova Joint and Muscle Relief TabletsTM (a tablet contains the water extract of SH) was approved by the Medicines and Healthcare products Regulatory Agency (MHRA) for the relief of backache, rheumatic pains, joint pain, and minor sports injuries, and become the first licensed traditional herbal medicinal product in the United Kingdom. The increasing worldwide acceptance of SH-related products greatly encourages the further research and development of SH.

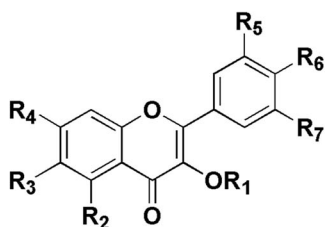
Phytochemicals

Up-to-date (2018), a total of 251 compounds have been isolated and identified from SO, SP and SG. Based on the chemical structures, these compounds could be divided into 6 groups: flavonoids, diterpenoids, sesquiterpenoids, triterpenoids, sterols as

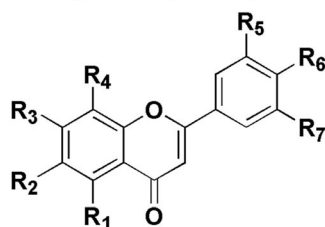
well as some others. The detail information of the reported compounds was summarized in the Supplement Table 1.

Flavonoids

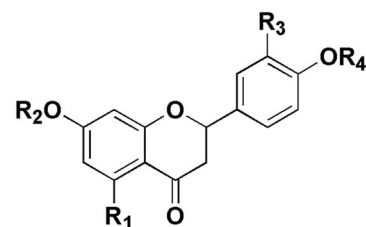
In total, 29 flavonoids have been isolated from SH species. The structures of the reported flavonoids were illustrated in Fig. 3. Among them, 3,7-dimethylquercetin (**15**) (Xu et al. 2015; Liu et al. 2012a; Lim et al. 2011; Kim et al. 2008; Xiong et al. 1997; Yang et al. 1976) was reported in all SH species, while rutin (**13**) (Yao et al. 2015; Phan 1999) and 3-methylquercetin (**14**) (Xu et al. 2015; Lim et al. 2011) were distributed in SO/SP or SP/SG, respectively. Specifically, two flavonols (3,7,4'-O-trimethylquercetin, **11** and 3,4'-O-dimethylquercetin, **12**) (Kim et al. 2008) were reported in SG, as well as 15 flavonoids including 10 flavonols (**1–10**) (Yao et al. 2015; Wang et al. 2014b), 9 flavones (**16–24**) (Yao et al. 2015; Wang et al. 2014b), 2 dihydroflavones (**25–26**) (Yao et al. 2015), 1



- 1^b R₁=Me R₂=OH R₃=H R₄=OH R₅=OMe R₆=OMe R₇=H
 2^b R₁=Me R₂=OH R₃=H R₄=OH R₅=OMe R₆=OH R₇=H
 3^b R₁=H R₂=OH R₃=H R₄=OH R₅=OH R₆=OH R₇=H
 4^b R₁=β-D-glc R₂=OH R₃=H R₄=OH R₅=H R₆=OH R₇=H
 5^b R₁=H R₂=OH R₃=H R₄=OH R₅=OMe R₆=OH R₇=H
 6^b R₁=β-D-glc R₂=OH R₃=H R₄=OH R₅=OH R₆=OH R₇=H
 7^b R₁=H R₂=OH R₃=H R₄=OH R₅=H R₆=OH R₇=H
 8^b R₁=α-L-rha R₂=OH R₃=H R₄=OH R₅=H R₆=OH R₇=OH
 9^b R₁=H R₂=OH R₃=H R₄=OH R₅=OH R₆=OH R₇=OH
 10^b R₁=H R₂=OH R₃=H R₄=OH R₅=OMe R₆=OMe R₇=H
 11^c R₁=Me R₂=OH R₃=H R₄=OMe R₅=OH R₆=OMe R₇=H
 12^c R₁=Me R₂=OH R₃=H R₄=OH R₅=OH R₆=OMe R₇=H
 13^{a, b} R₁=β-D-glc (1-2)-α-L-rha R₂=OH R₃=H R₄=OH R₅=H
 R₆=OH R₇=OH
 14^{b, c} R₁=Me R₂=OH R₃=H R₄=OH R₅=OH R₆=OH R₇=H
 15^{a, b, c} R₁=Me R₂=OH R₃=H R₄=OMe R₅=OH R₆=OH R₇=H



- 16^b R₁=OH R₂=H R₃=OMe R₄=H R₅=OMe R₆=OMe R₇=H
 17^b R₁=OH R₂=H R₃=OH R₄=H R₅=OH R₆=OH R₇=H
 18^b R₁=OH R₂=H R₃=OH R₄=H R₅=OH R₆=H R₇=H
 19^b R₁=OH R₂=H R₃=OH R₄=C-glc R₅=H R₆=OH R₇=H
 20^b R₁=OH R₂=H R₃=O-β-D-glc R₄=H R₅=H R₆=OH R₇=OH
 21^b R₁=OH R₂=OH R₃=OGlcA R₄=H R₅=H R₆=H R₇=H
 22^b R₁=OH R₂=β-D-glc R₃=OH R₄=β-D-glc R₅=H R₆=OH R₇=H
 23^b R₁=OH R₂=H R₃=OH R₄=β-D-glc (1-2)-α-L-rha
 R₅=H R₆=OH R₇=H
 24^b R₁=OH R₂=H R₃=OH R₄=β-D-glc R₅=H R₆=H R₇=H



- 25^b R₁=OH R₂=β-D-glc (1-2)-α-L-rha R₃=OH R₄=Me
 26^b R₁=H R₂=H R₃=H R₄=β-D-glc

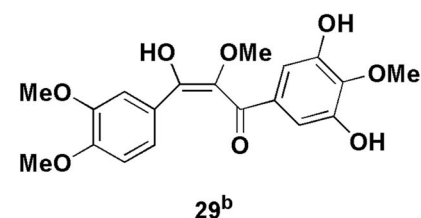
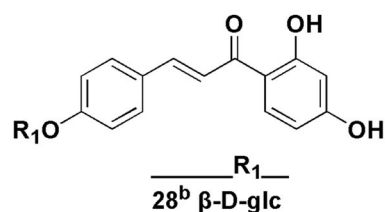
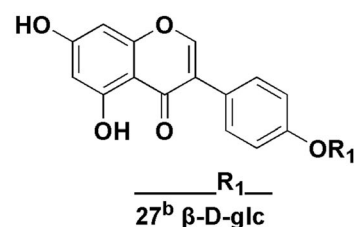


Fig. 3 The chemical structures of the flavonoids identified in *SigesbeckiaeHerba*. ^a*Sigesbeckia orientalis* L. (SO), ^b*S. pubescens* Makino (SP) and ^c*S. glabrescens* Makino (SG)

isoflavone (27) (Yao et al. 2015) and 2 chalcone derivatives (28–29) (Yao et al. 2015; Chou and Wang 2006) were reported in SP. As 24 out of 29 flavonoids

are flavonols (compound 1–15) or flavones (compound 16–24), it indicates that the SH-derived flavonoids might be mainly belonged to these two

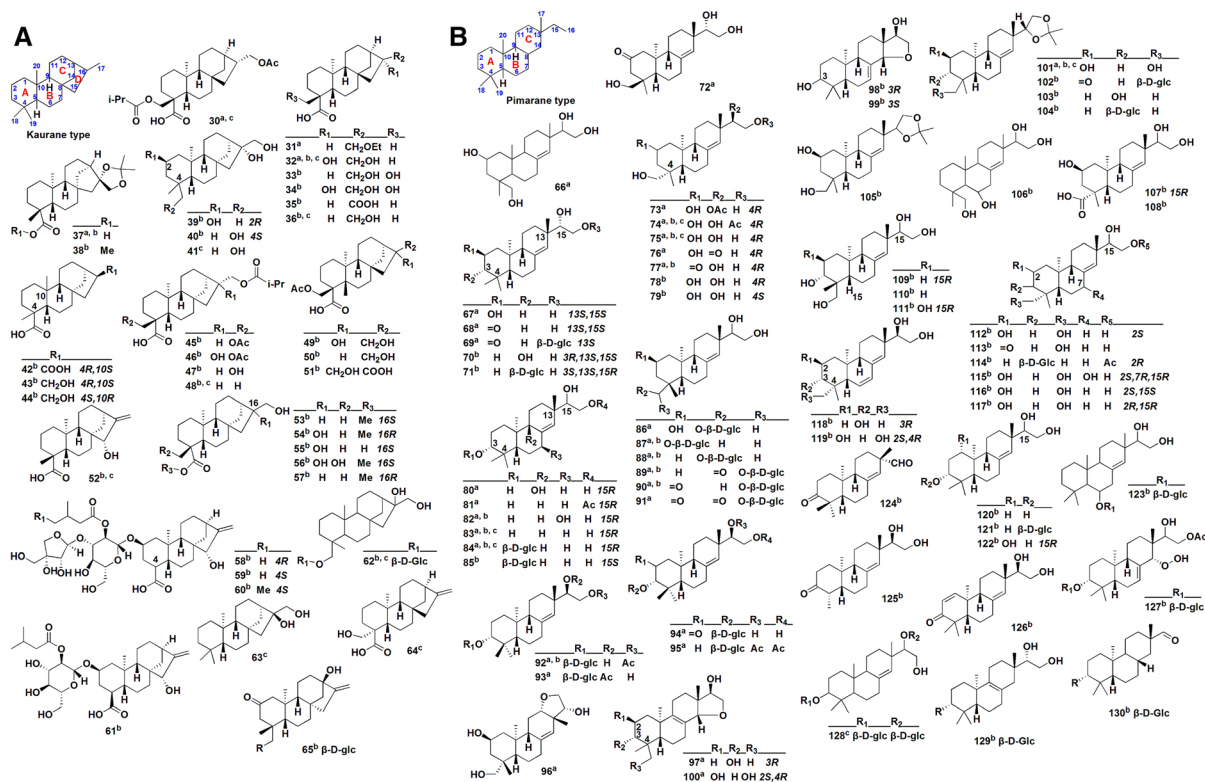


Fig. 4 The chemical structures of kaurane-type diterpenoids (**a**) and pimarane-type diterpenoids (**b**) identified in *Sigesbeckiae* Herba. ^a*Sigesbeckia orientalis* L. (SO), ^b*S. pubescens* Makino (SP) and ^c*S. glabrescens* Makino (SG). no indication color for SG on illustration

types. However, since most of these flavonoids were reported in SP while limitedly investigated and reported for SO and SG, it is hard to conclude the characteristics of flavonoids distribution in individual species at the current status. On the other hand, these flavonoids are also common compounds, which widely distributed in various plants. Therefore, future investigations need to be conducted to explore whether there are specific flavonoids in SH plants or individual species.

Diterpenoids

Diterpenoid is one of the most important types of chemicals in SH herbs. The representative chemical skeletons of SH diterpenoids are tetracyclic kaurane and tricyclic pimarane. The reported kaurane-/pimarane-type diterpenoids in SO, SP and SG were 4/29, 31/47 and 9/6, respectively. In addition, 2 strobane-type diterpenoids were also reported in SP.

Kaurane-type diterpenoids

The basic skeleton of kaurane-type diterpenoids (KDPs) is constituted by a perhydrophenanthrene unit (A, B and C rings) fused with a cyclopentane unit (D ring), which formed by a bridge of two carbons between C-8 and C-13 (Fig. 4a) (Garcia et al. 2007). These kaurane diterpenoids could be distinguished by their different substituents at C-16, C-17, C-18 and C-19, while few also found to be substituted at C-2. All the chemical structures of KDPs found in three SH species were illustrated in Fig. 4a.

Four KDPs with a carboxyl at C-19 were isolated from the aerial parts of SO, named as siegesetheric acid I (**30**), siegesesteric acid II (**31**), annosquamosin A (**32**) (Guo et al. 1997) and *ent*-16 β ,17-dihydroxykauran-19-oic acid-16 β ,17-acetonide (**37**) (Xiong et al. 1997), respectively. Meanwhile, compound **32** (Wang et al. 2009a, b; Fu et al. 1998a; Guo et al. 1997) was also found in SP and SG too; while compound **30** (Kim et al. 2006) and compound **37** (Wang et al.

2009a, b) were also reported in SG and SP. Earlier, there were three KDPs (**39–41**) without carboxyl moiety at C-19 found in SH. Siegesbeckiol (**39**) (Xiong et al. 1992) was substituted with trihydroxy at C-2, C-16 and C-17 on its kaurane-type diterpenoid carbon skeleton. *Ent*-kauran-16 β ,17,18-triol (**40**) (Xiong et al. 1992) and kaurane-16,17,18-triol (**41**) (Fu et al. 1998a) were a pair of stereoisomers. Similar to those in SO, most KDPs isolated from SP have presented a carboxyl moiety at C-19. As for SP-derived compound **38**, it could be distinguished from compound **37** by its substituted esterification at C-19 (Wang et al. 2009a, b).

There were three other kaurane-type diterpenoids isolated from SP showed structural similarities to compound **40**, namely *ent*-17,18-dihydroxy-16 β H-kauran-19-oic acid (**33**) (Lee et al. 2014a, b; Xiong et al. 1992), *ent*-16 α , 17,18-trihydroxykauran-19-oic acid (**34**) (Gao et al. 2002; Wang et al. 2009a, b) and siegeskaurolic acid (**36**) (Park 2006; Xiong et al. 1992). Compound **35** formed a carboxyl at C-17 of siegeskaurolic acid, (Wang et al. 2009a, b) and so was its stereoisomer, which named kauralexin A2 (**42**). (Han et al. 1975) Additionally, three other stereoisomers of siegeskaurolic acid were found in SP, which confirmed to be (-)-17-hydroxy-16 α -kauran-19-oic acid (**43**) (Wang et al. 2009a, b), *ent*-17-hydroxy-16 β H-kauran-19-oic acid (**44**) (Xu et al. 2015) and *ent*-16 α , 17-dihydroxy-kauran-19-oic acid (**55**) (Sung et al. 2011) respectively. Compound **45–48** not only shared a carboxyl at C-19, but also an isobutyrate group at C-17. And compound **49–51** not only shared a carboxyl at C-19, but also an acetic acid ester at C-18 (Wang et al. 2009a, b, 2014a). Compared with compound **49** and **50**, compound **51** had another carboxyl at C-17. Structurally, grandifloric acid (**52**) (Xiong et al. 1992) formed a carbon–carbon double bond between C-16 and C-17. The carboxyl at C-19 of compound **53–54** and **56–57** has been substituted by a methyl ester. Among them, compound **53** and compound **57** were stereoisomers of each other. In addition, four *ent*-kaurane glucopyranosides (**58–61**) (Liu et al. 2012b) have been isolated from the aerial parts of SP. Earlier, another *ent*-kaurane glucopyranoside named siegesbeckioside (**62**) (Xiong et al. 1992) was isolated. And recently, a new glycoside named siegeside E (**65**) was also found in SP (Wang et al. 2017b).

It needs mentioning that compound **36**, **48** (Kim et al. 2006), **56**, (Fu et al. 1998b) and **62** (Ma et al. 1998) have also been found in SG. Other kaurane diterpenoids found in SG were 16 β ,17-dihydroxy-kaurane (**63**) (Fu et al. 1998b) and (4 β)-19-hydroxy-kaur-16-en-18-oic acid (**64**) (Fu et al. 1998a). As 31 out of 35 KDPs have been found in SP, one consensus could be concluded after comparing with other compounds is that KDP might be a representative group of compounds of SH. However, the limited numbers of KDPs in SO and SG could be resulted from the lack of related researches. There is another possibility there KDP might also be a representative group of compounds for SP to distinguish from other SH species. Both hypotheses require further studies to confirm.

Pimarane-type diterpenoids

The pimarane-type diterpenoids (PDPs) share a common pimarane skeleton, which is constituted by a perhydrophenanthrene unit (A, B and C rings) with an angle methyl at C-4 position of A ring, and a methylethyl at C-13 position of C ring (Fig. 4b) (Porto et al. 2009). Their substitutions mainly occur at C-2, C-3, C-15, C-16, C-18 and C-19. Possible substitutions also have been found at C-6, C-7, C-14. Other differences on structure were the locations of carbon–carbon double bond within the perhydrophenanthrene skeleton. All the chemical structures of PDPs found in SH plants were shown in Fig. 4b.

The majority of PDPs found in SO formed a carbon–carbon double bond between C-8 and C-14 (**66–69**, **72–77**, **80–84**, **86–96**, **101**). Based on substitutions at C-2, compound **66–77** could be classified into two groups: hydroxyl and glycosylation substituted, respectively. Among them, compound **68–70**, and **77** have formed a carbonyl at C-2, while compound **69** and **77** were also stereoisomers of each other (Zdero et al. 1991; Xiang et al. 2004). Compound **113** is a stereoisomer of compounds **72** and **77**, which isolated from SP (Wang et al. 2014a). Another cluster of PDPs with a hydroxyl at C-2 were named as follows: *ent*-2 α ,15,16,19-tetrahydroxypimar-8(14)-ene (**66**) (Xiang et al. 2004), 2 β ,15,16-trihydroxy-*ent*-pimar-8(14)-ene (**67**) (Zdero et al. 1991), orientalin A (**73**) (Xiong et al. 1997), orientalin B (**74**) (Xiang et al. 2004), kirenol (**75**) (Wang et al. 2009a, b; Xiang et al. 2004) and *ent*-15-oxo-2 β ,16,19-

trihydroxypimar-8(14)-ene (**76**) (Xiang et al. 2004). Compound **112** is a stereoisomer of compounds **66** and **75**, which isolated from SP (Wang et al. 2014a). Three compounds (**80–82**) (Wang et al. 2009a, b) were found to be substituted hydroxyl at C-3. The hydroxylated position of compound **80** and **81** occurred at C-9, while compound **77** was at C-7. For darutigenol (**83**) (Wang et al. 2009a, b), compound **120** was a stereoisomer isolated from SP (Wang et al. 2014a). Five *ent*-pimarane glucopyranosides with glycosidic bond at C-3 were also isolated from SO, named as darutoside (**84**), hythiemoside A (**92**) and 15,16-di-*O*-acetyldarutoside (**95**) (Wang et al. 2009a, b), hythiemoside B (**93**) (Giang et al. 2005) and *ent*-2-oxo-3 β ,15,16-trihydroxy-pimar-8(14)-en-3-*O*- β -glucopyranoside (**94**) (Xiang et al. 2004), respectively.

Compound **114** and **121** were two *ent*-pimarane glucopyranosides isolated from SP, which were the stereoisomers of hythiemoside A and darutoside, respectively (Wang et al. 2014a). Six other *ent*-pimarane glucopyranosides (**86–91**) (Xiang et al. 2004) shared an ethylene glycol moiety at the side chain of C-13–16, which was quite common in pimarane diterpenoids. However, the substituted glycoside was at different position. Among them, only pubeside A (**87**) formed glycosidic bond at C-2, while others were at the side chain of C-18. A carbonyl was formed at C-2 of both pubeside D (**90**) and compound **91**. The glycosidic carbon of both pubeside C (**89**) and compound **91** was carbonylated. Through an oxygen bridge, an oxolane moiety was formed between C-12 and C-16 of compound **96** (Xiang et al. 2004). Differently, the oxygen bridge was formed between C-14 and C-16 of compound **97** and **100** (Wang et al. 2009a, b). The carbon–carbon double bond of these two compounds was formed between C-8 and C-9. For isopropylidenekirenol (**101**) (Xiang et al. 2004), a different location of carbon–carbon double bond was between C-8 and C-14, while a five -membered dioxygen heterocyclic moiety was formed at its C-15–16 side chain. Moreover, compound **77**, **82** and hythiemoside A (**92**) (Wang et al. 2009a, b), as well as pubeside A–D (**87–90**) (Xiong et al. 2001) were also found in SP. Whereas orientalin B (**74**) (Liu et al. 2012a; Xiang et al. 2004; Liu and Roder 1991), kirenol (**75**) (Wang et al. 2009a, b; Xiang et al. 2004; Liu and Roder 1991), darutigenol (**83**) (Wang et al. 2009a, b; Giang et al. 2005; Dong et al. 1989), darutoside (**84**) (Dong et al. 1989; Wang et al.

2009a, b) and isopropylidenekirenol (**101**) (Wang et al. 2009a, b; Xiang et al. 2004; Liu and Roder 1991) were common ingredients in all three SH plants.

Four other SP-derived PDPs compound **102–105** shared a common dioxo-5-membered ring moiety at C-15–16 side chain in their structure (Xiang et al. 2004; Wang et al. 2009a, b, 2014a). Among them, compound **105** was a stereoisomer of isopropylidenekirenol. Structurally, compound **106** was substituted by a hydroxyl at C-6 of compound **66** (Canonica et al. 1969). Compound **107** (Wang et al. 2009a, b) and **108** (Wang et al. 2014a) was a pair of stereoisomers, both isolated from SP. Likewise, compound **109** (Wang et al. 2009a, b) and **110** (Wang et al. 2014a) was a pair of stereoisomers, also isolated from SP. Compound **123** was uniquely substituted glycoside at C-6 (Kim 1972). Carbonylation and aldehyde substitution were occurred at C-3 and C-13–16 side chain of compound **124** (Xie et al. 2013). Different from other PDPs found in SP, compound **127** formed a carbon–carbon double bond between C-7 and C-8. However, as the hydrogen peroxide substitution at C-14 of compound **127** could be unstable, it might undergo an allylic rearrangement followed by a slower Smith epimerization to another hydroperoxide (Wang et al. 2009a, b). Recently, series of PDPs with a common carbon–carbon double bond between C-8 and C-14 in their structures were isolated from SP, such as siegeside B aglycon (**70**), siegeside A (**71**), compound **78–79**, siegeside B (**85**), compound **98–99**, **111**, **115–117**, and **122** (Wang et al. 2017a). Compound **118–119** were also found in SP, but formed two carbon–carbon double bonds in their structures. One was located at C-6 and -7, the other was at C-8 and -14. There were two different diterpene glycosides found in SP. One is siegeside C (**129**) with a carbon–carbon double bond between C-8 and -9, and the other is siegeside D (**130**) with an aldehyde group at C-15 position (Wang et al. 2017a). Additionally, neodarutoside (**128**) was isolated from SG, which had two glucopyranoside subunits at C-3 and C-15 respectively (Dong et al. 1989).

It should be mentioned that a pimarane diterpenoid isolated from SO was firstly confirmed as *ent*-12 α ,16-epoxy-2 β ,15 α ,19-trihydroxypimar-8-ene (Xiang et al. 2004), but revised as compound **100** later (Wang et al. 2009a, b). Darutin was also isolated from SO, which remaining incompletely defined yet. And it is also worth noting that two *ent*-strobane diterpenoids have

been isolated from SP recently, named strobols A (**131**) and strobols B (**132**) respectively (Wang et al. 2017a). A total of 65 PDPs have been found in SH species, indicating PDP might be another representative group of compounds of SH. Up-to-date, there are 29 PDPs existing in SO, 47 in SP, while only 6 in SG. Compared PDPs distribution in SO and SP, the structural difference is not significant. Perhaps the content of PDPs differs among species, which needs future comparative studies to address. And the chemical composition illumination of SG is insufficient, which also needs further investigations.

Sesquiterpenoids

The number of sesquiterpenoids reported in SO, SP and SG were 35, 26 and 4, respectively. These sesquiterpenoids could be classified into germacranolides, cadinane and other types. All the chemical structures of sesquiterpenoids found in SH plants were shown in Fig. 5.

Compounds **133–164** shared a similar carbon skeleton constituted of a germacrane ring fused with a heterocyclic, formed by γ -lactonization between C-8 and C-12. And also a carbon–carbon double bond between C-11 and C-13 was in common (Fig. 5). Possible substitutions would occur at C-5, C-6, C-14 and C-15. And the position of carbon–carbon double bond within the germacrane ring would also be differentiated. Among them, compound **133–141**, **145–154** and **157–164** were isolated from SO (Wang and Hu 2006; Xiang et al. 2005; Zdero et al. 1991), compound **142–143** were isolated from SP (Wang et al. 2014a), while compound **155–156** (Wu et al. 2015; Li et al. 2011) were isolated from SG. A germacranolid named as pubetallin (**144**) (Wang and Hu 2006; Kim and Ahn 1988) was found in both SP and SG. Uniquely, an oxirane moiety between C-3 and C-4 was formed among compound **161–164** (Zdero et al. 1991). Compound **165–166** were two other germacranolides identified in the leaf oil of SP, which without heterocyclic subunit in structure (Sakuda 1987).

Series of cadinane sesquiterpenoids (**167–172**) were found only in SP, distinguished by different number and location of carbon–carbon double bonds. Compound **167–170** were identified in leaf oil of SP (Sakuda 1987), while compound **171–172** were identified from volatile oil in the stem and leaf of SP (Gao

et al. 2000). Moreover, compound **167**, **168** and compound **169**, **171** were two pairs of stereoisomers, respectively.

Two eudesmane sesquiterpenoids, namely α -copaene (**173**) and $1\beta,6\alpha$ -dihydroxy-4(14)-eudesmene (**174**) were found in SP (Sakuda 1987). Spathulenol (**175**) (Sakuda 1987) and (+)-4a-allospathulenol (**176**) (Gao et al. 2000) were two aromadendrane sesquiterpenoids also found in SP. Three guaianolides compound **177–179** were isolated from SO (Xiang et al. 2005). Earlier, two eudesmanolides compound **180–181** were also isolated from SO (Zdero et al. 1991). In addition, two sesquiterpene lactones, namely as siegenolide A (**182**) and siegenolide B (**183**), were isolated from SG together with compound **155–156** (Wu et al. 2015). Three other sesquiterpene lactones compound **184–186** were isolated from SP (Wang et al. 2014a). Other sesquiterpenoids (compound **187–194**) found in SP were listed in Supplement Table 1 (Wang et al. 2014a).

As sesquiterpenoids found in SH plants are common compounds, it might not be a representative group of compounds for SH. However, as germacranolides mainly occur in SO, while cadinane sesquiterpenoids mainly occur in SP, it suggests that sesquiterpenoids might be suitable chemical compositions to distinguish different SH species.

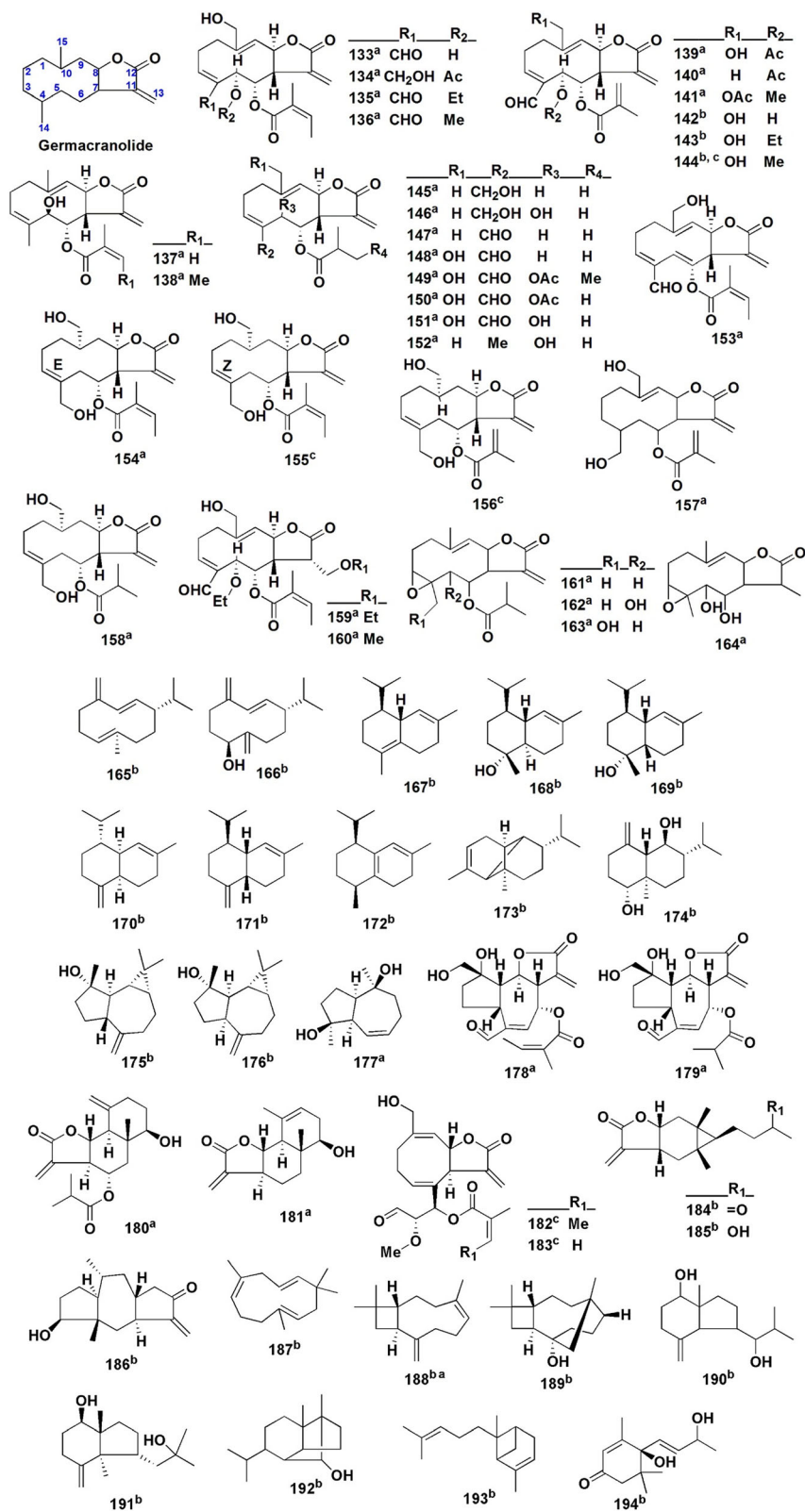
Other compounds

Three ursane-type pentacyclic triterpenoids (Wang et al. 2014b) and six steroids (Xiong et al. 1992; Wang et al. 2014b; Jiang et al. 2009) have been isolated from SP, including β -sitosterol (**198**) and β -daucosterol (**199**), which have also been found in SO and SG (Fu et al. 1998a; Guo et al. 1997). Other fat-soluble compounds found in SH species have been summarized in Supplement Table 1. These compounds are very common among different plants, which might not be meaningful for the quality control of SH. However, their bioactivities should not be ignored.

Phytopharmacological effects

Multiple biological activities for different SH extracts and SH-derived compounds have been investigated and reported, including anti-inflammation, immunomodulation, anti-hypertension, anti-platelet,

Fig. 5 The chemical structures of sesquiterpenoids identified in *Sigesbeckiae Herba*.
^a*Sigesbeckia orientalis* L. (SO), ^b*S. pubescens* Makino (SP) and ^c*S. glabrescens* Makino (SG)



anti-thrombotic, anti-cancer, anti-bacteria and other effects. An overview on the present status of pharmacological effects for SH extracts and SH-derived compounds has been summarized in Supplement Tables 2 and 3 respectively, which based on the method of extraction, quality control, and administration.

Anti-inflammation

Anti-inflammation is one of the most important activities for SH, which is its main traditional clinical application. Reported by Hong et al. pre-treatment of the 95% ethanol extract of SO (95EESO) (10 and 32 mg/kg, daily) for 1 week was observed to significantly attenuate the λ -carrageenan (s.c.)- or LPS (i.p.)-induced acute inflammation (degree of paw edema, serum cytokine levels, and survival rate) in mice model dose-dependently. However, compared with positive control indomethacin (s.c., 20 mg/kg) in λ -carrageenan model, the effect of 95EESO was lower than half effect of indomethacin. While in LPS model, 95EESO exhibited relatively similar effect to the positive control PDTC (i.p., 50 mg/kg). As λ -carrageenan and LPS are used to simulate local acute inflammation and systemic inflammation respectively, it suggests that 95EESO might respond effectively to specific inflammation. Moreover, the biological mechanisms of 95EESO on anti-inflammation were related to the suppression of MAPKs- (ERK1/2, p38, and JNK phosphorylation) and NF- κ B-dependent pathways. Additionally, four active ingredients (**250**: caryophyllene oxide; **175**: [-]-spathulenol; **251**: hexadecanoic acid ethyl ester; and **75**: kireinol) were detected to standardize the extract, among which kireinol occupied 0.412–0.428%. (Hong et al. 2014).

The ethanoilc and methanolic extracts of SP have been reported to present various therapeutic effects in collagenase-induced (CIA) arthritic diseases. In a rabbit model with CIA-induced osteoarthritis, the 50% ethanol extract of SP (50EESP) (p.o., 100, 400 mg/kg), which used *ent*-16 α H,17-hydroxy-kauran-19-oic acid ($1.36 \pm 0.03\%$) for quality control, was observed to protect CIA-induced cartilage destruction by down-regulating the inflammatory mediators in synovial fluid, and the expressions of aggrecanases and MMPs in cartilages and synovial tissues of knee joints. Moreover, the up-regulation of the TIMP-1 expression in cartilages and synovial tissues was also determined

in 50EESP-treated rabbits. Compared with positive control celecoxib (p.o., 100 mg/kg), both dosages of 50EESP were more effective to relieve stiffness of rabbit knee joints, indicating the potential of 50EESP to treat arthritis (Huh et al. 2008). Recently, a 50% ethanol extract of SP with different extraction process was investigated to inhibited Pam3CSK4-induced inflammation in RAW 264.7 macrophages through suppressing TLR1/TLR2-mediated NF- κ B activation (Sang et al. 2018). Due to lack of animal experiment verification, it's hard to compare its activity with 50EESP, and decide whose extraction method is better. As it has chosen different ingredients (**13**: rutin, $0.27 \pm 0.01\%$; **75**: kireinol, $1.81 \pm 0.02\%$; **84**: darutoside, $0.28 \pm 0.03\%$) for quality control, it's also unable to compare their chemical components. Ointments containing 1–5% methanolic extract of SP could effectively reduce the carrageenan-induced edema and presented analgesic activity in rats (Wang et al. 2008). Furthermore, by comparing the anti-nociceptive and anti-inflammatory activities of methanolic extracts of various SP samples (unprocessed and processed) in mice and rat, a bioactive diterpenoid named siegeskaurolic acid (**36**) was finally demonstrated and isolated (Park 2006). Neither of these two methanolic extracts provided their quality control methods. Another concern is methanolic extract might not be suitable for industrial production, while ethanoilc extract has been considered safer and practical. Moreover, pre-treatment of 50% ethanol extract of SO (0.75 or 1.5 g/kg/day, oral) was found to attenuate postoperative cognitive dysfunction, systemic inflammation, and neuroinflammation in mice, suggesting the therapeutic potential of *S. Orientalis* in minimizing POCD and the possibility of utilizing this traditional Chinese medicine perioperatively (Chu et al. 2018).

In addition, the methanolic extract of SG was observed to attenuate allergic airway inflammation in LPS-stimulated RAW264.7 cells and ovalbumin (OVA)-induced asthma mice with the potential mechanisms on suppressing inflammatory responses via the reduction of iNOS and COX-2 expression in lung tissue (Jeon et al. 2014). Though the anti-asthma effect of this SG extract (oral, 30 mg/kg, daily) was comparatively close to the positive control montelukast (oral, 30 mg/kg), its extraction and quality control method are not given.

Meanwhile, various compounds in SH species have been reported to present anti-inflammatory effects both in vitro and in vivo. Among its constituents, terpenoids and flavonoids are two most important types of compounds in SH herbs which contributed to their anti-inflammatory activities. Four quercetin derivatives (**11**: 3,7,4'-*O*-trimethylquercetin; **12**: 4'-*O*-dimethylquercetin; **14**: 3-methylquercetin and **15**: 3',7-dimethylquercetin) isolated from SG were reported to effectively inhibit the nitric oxide (NO) production in LPS activated microglia (Lim et al. 2011; Kim et al. 2008). All of them could reduce the protein and mRNA expressions of cyclooxygenase-2 (COX-2) and inducible nitric oxide synthase (iNOS) through suppressing the activation of nuclear factor kappa-light-chain-enhancer of activated B cells (NF- κ B), as well as the phosphorylation of p38 and extracellular signal-regulated kinases (ERK). Moreover, **14** and **15** exhibited scavenging activity for peroxynitrite. Apparently, flavonoids are important active compounds in SH plants. Whereas their contribution to the bioactivity of SH plants depend on their contents. Thus, it's necessary to address the content of total flavonoids in SH plants, which hasn't been valued in any reports yet.

It's a consensus that diterpenoids are not only chemically representatives of SH plants, but also the material basis of activity. Seven of kaurane-type (**33**, **36**, **47**, **48**, **54**, **55** and **56**) and one pimarane-type (**75**) diterpenoids were reported to inhibit the LPS-induced over-production of NO in BV2 microglia (Lee et al. 2014a, b). The anti-inflammatory effects of siegeskaurolic acid (**36**) (Park et al. 2007) and kirenol (**75**) (Lee et al. 2014a, b) were further identified to be associated to the decrease of iNOS and COX-2 expressions via NF- κ B inactivation. Moreover, intragastric administration of siegeskaurolic acid in mice and rats (20 and 30 mg/kg/day) presented anti-nociceptive and anti-inflammatory effects by hot plate- and writhing anti-nociceptive assays and carrageenan-induced anti-inflammatory assays (Park 2006).

The topical anti-inflammatory and analgesic effects of kirenol have been confirmed in carrageenan-induced acute inflammation, complete Freund's adjuvant (CFA)-induced chronic inflammation and formalin test in rats. The anti-inflammatory activity of kirenol 0.4 ~ 0.5% (w/w) was investigated to be similar to that of piroxicam gel after carrageenan injection. Moreover, topical administration of kirenol

cream at the dose of 0.3, 0.4 and 0.5% (w/w) could significantly inhibit the development of joint swelling induced by CFA (Wang et al. 2011b). Reported by Wang et al. in collagen-induced arthritis (CIA) rat model, kirenol (1, 2, and 4 mg/kg) could depress paw swelling and reduce IL-1 β of synovial fluid, by upregulating nuclear Annexin-1 and inhibiting NF- κ B activity in synovium of CIA (Wang et al. 2011a, b, c, d). These therapeutic effects on CIA might be due to its modulation on T cells balance (Lu et al. 2012).

Reported by Li et al. a sesquiterpene lactone found in SG (compound **156**) presented inhibitory effects on NO and PGE2 synthesis in LPS-induced RAW 264.7 cells. Moreover, the protein and mRNA levels of iNOS and COX-2 could also be dose-dependently decreased by compound **135** (Li et al. 2011). In addition, siegeskaurolic acid was investigated to attenuate LPS-induced NF- κ B by inhibiting the nuclear translocation of the p65 subunit of NF- κ B via blocking the I κ B- α degradation (Park et al. 2007).

Other compounds, such as ursolic acid (**195**), β -sitosterol (**198**) and tetracosacarbonic acid (**235**), could also inhibit the NO production from LPS-induced RAW 264.7 cells, while ursolic acid could also inhibit the release of TNF- α (Zhao et al. 2012).

Immunomodulation

The water extract of SO was reported to present anti-allergic activity both in vitro (in LPS plus IL-4-activated U266B1 human IgE-bearing B cells) and in vivo (in antigen-induced mice) by inhibition of immunoglobulin E (Ig E) production from B cells dose-dependently (in vitro, 0.1, 1 mg/ml; in vivo, p.o. or i.p., 0.1–1000 g/kg). (Hwang et al. 2001) However, there are several concerns about this study. Though in vitro screening has shown considerable positive results, the in vivo dosage in this study is suspicious. At such high dosage might cause side effect or toxicity. The 70% ethanol extract of SO (70EESO), which chosen kirenol (1.04 $\pm$ 0.02%) for quality control, was reported to present good immunosuppressive and anti-inflammatory activities. Intraperitoneal administration of 70EESO (0.25, 0.5, and 1.0 mg/0.2 mL per hind foot at intervals of 7 days) dose-dependently suppressed concanavalin A (ConA)-, lipopolysaccharide (LPS)- and ovalbumin (OVA)-induced splenocyte proliferation, and OVA-

specific serum IgG, IgG1, and IgG2b levels in OVA-immunized (s.c.) mice (Sun and Wang 2006). It is promising that 70EESO at the highest dosage (1.0 mg/0.2 mL) even showed better results than the positive control cyclosporin A (0.1 mg/0.2 mL). And according to the content of kireinol, the extraction efficiency 70EESO is higher than 95EESO (Hong et al. 2014), indicating 70% ethanol might be better solvent for extraction.

Anti-allergic activities were widely reported for SG. The water extract of SG was observed to effectively improve systemic anaphylaxis both in vitro and in vivo, which mainly due to the inhibition of histamine release from mast cells (Kang et al. 1997) and IgE production from B cells (Kim et al. 2001). Though the 70 °C water extract of both SO and SG exhibited anti-allergic effect, the dosage of SG water extract in the study is more reasonable, which might also suggest SG has more potential than SO. However, as BALB/c mice (compound 48/80-induced systemic anaphylaxis) and ICR mice (active systemic anaphylaxis) have been orally administered with SG water extract at the dosages of 0.1–10,000 and 50–1000 mg/kg respectively, whilst Wistar rats (passive cutaneous anaphylaxis) was administered at the dosage of 100 mg/kg, it is confused that how BALB/c and ICR mice with stand higher dosages than Wistar rats with smaller body size.

Anti-cancer

Reported by Chang et al., 95EESO effectively inhibited the proliferation of various cancer cells in vitro including RL-95 (endometrial cancer), A549 (lung cancer), Hep G2 (hepatoma), FaDu (pharynx squamous cancer), MDA-MB-231 (breast cancer), and LNCaP (prostate cancer) cells (Chang et al. 2014). The anti-cancer properties of 95EESO were investigated to be associated to the modulations of G2/M cell arrest and expressions of apoptosis-related proteins. The related bioactive constituents were further identified to be caryophyllene oxide (**250**), caryophyllene (**188**), 6,10,14-trimethyl-2-pentadecanone (**228**), and hexadecanoic acid ethyl ester (**251**), indicating fat-soluble components couldn't be ignored in SH-related researches.

The anti-cancer activities of SG have been reported to MCF-7 and MDA-MB-231 cells (breast cancer), SKOV-3 cells (ovarian cancer), A549 and H1299 cells

(non-small cell lung cancer). A water extract of SG could induce apoptosis in both MCF-7 and MDA-MB-231 cells, marked by cleavages of poly (ADP-ribose) polymerase (PARP) and different pre-caspases (Jun et al. 2006). An ethanol extract of SG exhibited inhibitory effects on proliferation, adhesion, migration and invasion of SKOV-3 cells, which mediated by suppression of FAK, ERK, Akt and p70^{S6K} dependent signaling pathways and down-regulation of receptor tyrosine kinases such as EGFR, VEGFR-2 and FGFR-1, as well as the cell adhesion molecule N-cadherin (Cho et al. 2013). In another report, the ethanol extract of SG inhibited the proliferation and invasion in both p53 wild-type A549 and p53-deficient H1299 cells, indicating a p53-independent manner (Lee et al. 2014a, b).

Meanwhile, several constituents from SH species have also shown their anti-cancer activity. A cluster of diterpenoids (**33**, **45**, **55**, **75**, **83**, **84** and **103**) (Sung et al. 2011) had been screened for their cytotoxic potential against HSC-T6, HeLa, and B16 cells. Among them, compound **45** and **95** showed moderate cytotoxic activity (Sung et al. 2011). In another report, kireinol showed strong cytotoxic effects on K562 cells by inducing apoptosis in a caspase-independent manner (Lu et al. 2014). Four sesquiterpenoids isolated from SG (compounds **156**, **157**, **182** and **183**) showed considerable cytotoxic activities against various cancer cells (MCF-7, AsPC-1, SW480, HCT 116, HepG2 and HeLa cells) (Wu et al. 2015). Moreover, the cytotoxic sesquiterpenoid namely pubetallin (**144**) against L1210 cells was isolated from SO and SP (Kim and Ahn 1988), and a cytotoxic carbamate (**236**) against A549 cells was isolated from SP (Liu et al. 2012a). These active components might contribute to the anti-cancer activity of SH species. However, all these in vitro results need to be further confirmed by in vivo studies in the future. And as SH species have no tradition of anti-cancer clinical applications, it's still not sure whether it's worth going deeper in this direction.

Other effects

The methanolic extract of SP exhibited wound healing properties to stimulate the growth of mouse fibroblast NIH3T3 (0.5–100 µg/mL) in vitro, and increases in the rate of wound contraction, period of epithelialization and content of hydroxyproline in excision wound

rats (topical, 3–5% SP ointments) in vivo (Wang et al. 2011a). In a deep venous thrombosis model of rats, an active fraction derived from the 70% ethanol extract of SP (80 and 160 mg/kg, i.g.) significantly decrease the serum levels of von Willebrand factor (vWF) and intercellular adhesion molecule-1 (ICAM-1) thus protecting vascular endothelial cells from derangement of venous thrombosis-induced inflammation (Wang et al. 2012a, b). Compound **55** is an SP-derived kaurane diterpenoid, which could increase the proliferation and migration of keratinocyte stem/progenitor cells (KSCs) by activating Akt/ERK and epithelial growth factor receptor (EGFR) pathways. Moreover, treatment with compound **55** could significantly induce the establishment of fully stratified epidermis and increased numbers of p63-positive cells in 3D culture model of KSCs, and successfully accelerated healing of epidermal wounds created by laser and dermatome in animal models (Sung et al. 2011). Reported by Park et al. compound **55** also showed the ability for activation of vasculogenic progenitor cells (VPCs) in a dose-dependent manner, with the increases of phospho-ERK1/2, and -Akt signals. To stabilize the phosphorylated EGF receptor of KSCs, the cytoplasmic protein tyrosine phosphatase was inhibited by compound **55**, which was not in VPCs, indicating a cell-type dependent manner. However, the compound did not increase the proliferation and tube formation in human umbilical vein endothelial cells (HUVECs) and human aorta endothelial cells (HAECs) as reported (Park et al. 2011). Additionally, there are a cluster of compounds isolated from SP, which are structurally similar to compound **55**, such as compound **53–54**, **56–57**. Their bioactivities should be investigated and compared in the future. As these compounds might form into a synergistic chemical group, which responsible for the wound healing, and anti-thrombotic activities of SP, their contents in SP as well as other SH species should also be investigated and compared in the future.

An ethanolic extract of SG was observed to present anti-angiogenic effects in VEGF-A-stimulated endothelial cell proliferation by suppressing the Akt and p70S6K-dependent signaling pathways (Kim et al. 2015). In addition, in the model of the isolated rat thoracic aorta, an ethanolic extract of SG (QC: kirenol, 0.67%) effectively inhibited both noradrenaline bitartrate (NA)- and potassium chloride (KCl)-induced contraction of endothelium-intact

aortic rings in an endothelium-independent pathways (Lee et al. 2013). Again, these in vitro studies need further in vivo studies to confirm.

Earlier, a common kaurane diterpenoid found in all SH species, namely annosquamosin A (**32**) has been demonstrated to present anti-hypertensive, anti-platelet and anti-thrombotic activities both in vitro and in vivo. In diphosphate (ADP)-, thrombin-, arachidonic acid-activated rat platelets, annosquamosin A was determined to significantly inhibit the platelet aggregation. And in rat model, annosquamosin A significantly reduced the thrombus weight and prolonged the prothrombin time (PT) and activated partial thromboplastin time (APTT) after intravenous injection (2.5–20 mg/kg) for successive 5 days. Furthermore, the underlying mechanisms of annosquamosin A on anti-platelet and anti-thrombosis were demonstrated to be related to the anticoagulatory effect and cAMP induction (Wang et al. 2011a, b, c, d; Han et al. 1975). Several structural similarities of annosquamosin A have been found in SH species, such as compound **31**, and **33–36**, indicating there might be synergistic chemical group. And the chemical composition and bioactivities of these compounds should be investigated in the future.

Except for the above-mentioned bioactivities, kirenol (**75**) was also reported to inhibit adipocyte differentiation and lipid accumulation through activation of the Wnt/ β -catenin signaling pathway in 3T3-L1 adipocytes (Kim et al. 2014b). In another report, kirenol was found to promote osteoblast differentiation through activating BMP and Wnt/ β -catenin signaling pathways in MC3T3-E1 cells (Kim et al. 2014a). And kirenol also exhibited anti-bacterial activities against gram-positive pathogens including *Staphylococcus epidermidis*, *Staphylococcus aureus*, and *Acinetobacter baumannii* (Wang et al. 2012a, b). Compound **237**, another compound isolated from SG, was screened to be specific for Gram-positive bacteria (*B. subtilis*, *E. faecalis*, *P. acnes*, *S. epidermidis*, *S. schleiferi* subsp. coagulans, *S. agalactiae* and *S. pyrogens*) with the minimal inhibitory concentrations (MIC) in the range of 3.12–25 mg/mL (Kim et al. 2012). Using the in vitro bioassay-guided fractionation approach, two PTP1B inhibitors named siegesesteric acid I (**30**) and *ent*-16 β H,17-isobutyryloxykauran-19-oic acid (**48**), were isolated from SG extract (Kim et al. 2006).

Toxicology

In ancient Chinese Medicine books, SH were recorded to be “small poison”. Although no significant clinical toxicity/side effects have been reported to SH, in recent years, some potential toxicities for the various SH extracts and SH-derived compounds were investigated and reported.

The acute and sub-acute toxicity of water decoction of SO in mice has been compared and reported (Guan et al. 2007). The median lethal dose (LD₅₀) for oral acute toxicity of SO water decoction in mice was determined to be 146.7 g (herb)/kg. Although no significant changes were observed for mice with 2-week consecutive oral administration, the pulmonary toxicity was observed while the dosage higher than 22.6 g (herb)/kg. Further compare to the reported LD₅₀ (414.3 g (herb)/kg) (Guan et al. 2005) of the alcohol precipitate from the water extract of SH, the toxicological compounds were suggested to be presented in the aqueous soluble part of SH extract. This has been further investigated and confirmed by comparing the toxicities of different extracts and fractions of SH in mice (Guan et al. 2012, 2009, 2008).

Moreover, the acute oral toxicity of annosquamosin A (**32**) (Wang, J. et al. 2011) and 50% ethanol extract of SP (Huh et al. 2008) were investigated and reported to be observed no significant toxic symptoms to experimental animals.

In addition, the comparative toxicities among different SH species were also investigated and reported (Gao et al. 2008). SO (LD₅₀: 39.44 g (herb)/kg) presented a little higher acute toxicity than that of SP (LD₅₀: 43.93 g (herb)/kg), but without significant difference.

Since the toxicological studies of SH, especial SG, are still inadequate, further investigations on the toxicity, as well as the elucidation on the potential toxic compounds and the related mechanisms need to be done to ensure safety of SH in clinical applications.

Discussion and perspectives

Although the three *Sigesbeckia* plants (SO, SP and SG) are recommended to be used for SH by the current Chinese Pharmacopeia (2015 Edition), the chemical constituents in SO, SP and SG have been more and more noticed and investigated in recent years (Tao

et al. 2018; Zhang et al. 2019). Based on the compounds obtained from the literature, the characteristic distribution of the main compounds in *Sigesbeckia* herbs were analyzed and illustrated in Fig. 6a, b. Among the reported compounds, diterpenoids, sesquiterpenoids and flavonoids were the main compounds for all three SH plants. The numbers of overlapping distributed compounds in SO/SP, SP/SG and SG/SO were 12, 6 and 1, respectively. Moreover, 9 compounds (namely 3',7-dimethylguercetin, annosquamosin A, orientalin B, kirenol, darutigenol, darutoside, isopropylidenekirenol, β -sitosterol, and β -daucosterol) were simultaneously existed in SO, SP and SG. From the perspective of similarity, these 28 common compounds might be considered and selected as the potential chemical markers for the qualitative analysis (fingerprinting) of the SH, although they have different content in each species. Other compounds with detectable concentration(s) which specifically distributed in one or two species could be considered as characteristic markers for their discrimination. Based on these strategies, we recommended 6 compounds based on a metabolomics analysis (namely *ent*-18-acetoxy-17-hydroxy-16 β H-kauran-19-oic acid, *ent*-18-acetoxy-16 α ,17-dihydroxykauran-19-oic acid, (6*S*,7*E*)-6,9-dihydroxymegastigma-4,7-dien-3-one, 7, 3',4'-trimethoxyluteolin, quercetin, and (4 β)-19-hydroxy-kaur-16-en-18-oic acid) as potential chemical markers for SH plant discrimination (Tao et al. 2018). The results indicated the distinguished chemical difference among SH species. Moreover, further studies for comprehensive comparisons in chemistry for SO, SP and SG should be conducted for better understanding their chemical bases related to their pharmacological activities and clinical applications.

Recent pharmacological studies demonstrated various biological and pharmacological activities to SH (Fig. 6c). The findings of anti-inflammatory and immunomodulatory activities of SH validated its well-known traditional and ethnopharmacological uses in the treatment of inflammatory diseases. However, current studies are insufficient to elucidate the potential pharmacological mechanisms on anti-inflammation for SH. Hence, further in-depth studies regarding relationship between the biological mechanisms and clinical efficacy of SH is essential. Moreover, since the chemical bases among SO, SP and SG are different, comparative comprehension on the detail pharmacological similarities and differences

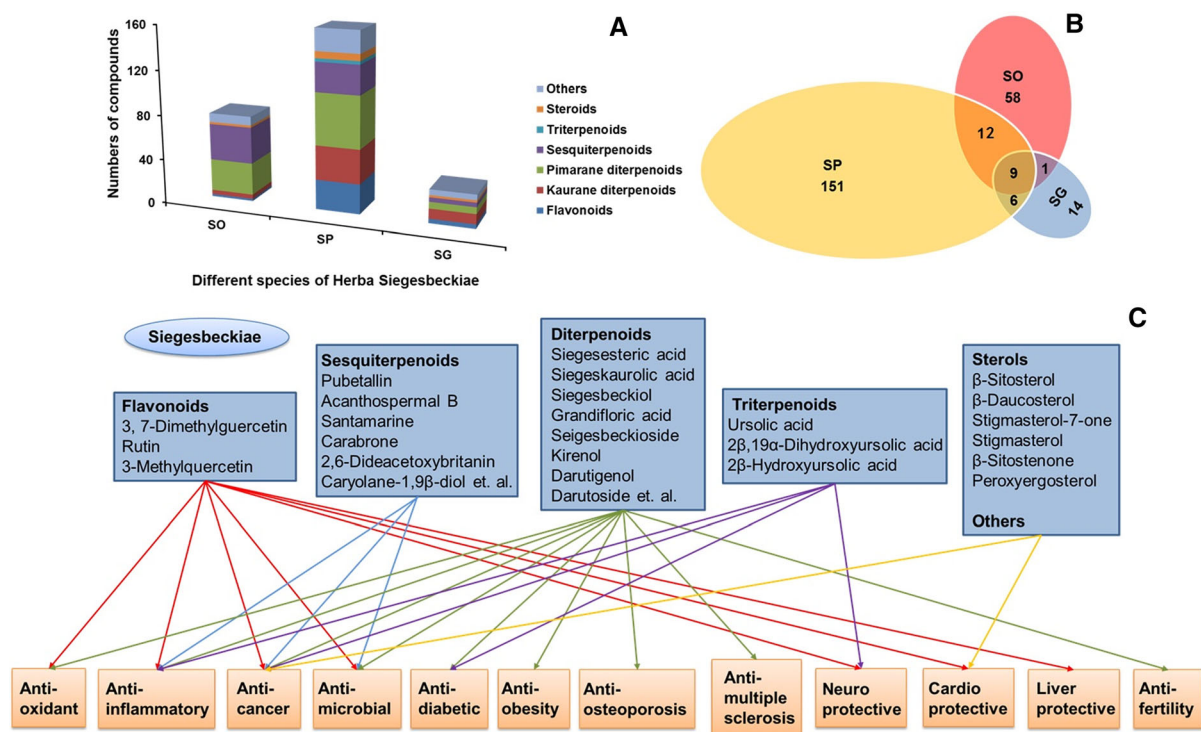


Fig. 6 Comparative understanding the chemical and pharmacological properties of *Sigesbeckia* plants. **a** Distribution of chemo-types in *Sigesbeckia* plants; **b** Characteristic distributions of chemical constituents in *Sigesbeckia* plants;

c Relationships between ingredients and bioactivities of *Sigesbeckia* plants. ^a*Sigesbeckia orientalis* L. (SO), ^b*S. pubescens* Makino (SP) and ^c*S. glabrescens* Makino (SG)

among SH species should also be conducted to improve our understanding for the quality control and precise clinical uses of these three *Sigesbeckia* species.

To be a traditional herbal medicine, the clinical efficacy of SH on inflammatory diseases have been long-term (over one thousand years in China) observed and identified. SH has been successfully applied for management or treatment of rheumatoid arthritis (Xie and Li 2015; Dou 2004), gouty arthritis (Yu et al. 2015; Zhou 2009), acute enteritis (Shi 2012), and chronic pains (Lin et al. 2019). Based on the sufficient evidence for demonstrated use of SH aerial parts for at least 30 years (of which at last 15 years in an EU member State), in 2015, the SH-derived traditional herbal medicinal product (Phynova Joint and Muscle Relief TabletsTM) has been approved by the Medicines and Healthcare products Regulatory Agency (MHRA) in UK. However, the used SH species in such pharmaceutical preparation still includes SO, SP and SG. Since the difference on

chemicals among the three SH plants were more and more investigated and reported, further comparison of the biological mechanisms and therapeutic effects of SO, SP and SG on anti-inflammation should be well conducted for their comparative and precise applications for rheumatism/-related diseases in clinics.

In recent years, *Sigesbeckiae* Herba has been more and more investigated with a rising attention by many researchers for their phytochemistry, pharmacology and toxicology. Owing to the fact that SH is originated from three plant species, comprehensive understanding of the *Sigesbeckia* herbs will renew our knowledge for this herbal medicine. From what have been discussed above, the collected information reviewed here exerts as a resource for future studies and medicinal exploitations of *Sigesbeckia* plants.

Acknowledgements This work was financially supported by the National Natural Science Foundation of China (NSFC, No. 81470170); the Science and Technology Development Fund, Macau SAR (No. 0096/2019/A2); and the Research Committee of the University of Macau (MYRG2017-00178-ICMS,

MYRG2018-00043-ICMS, SRG2015-00060-ICMS-QRCM and CPG2019-00006-ICMS).

Author contributions YH and WYT organized, conceived, and supervised the study; THX, XW, SW and PY performed the literature collection; THX, LKG, ZGD and PY performed data analysis; THX drafted the manuscript; THX and YH edited and revised the manuscript. All authors have read and approved the final manuscript.

Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

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